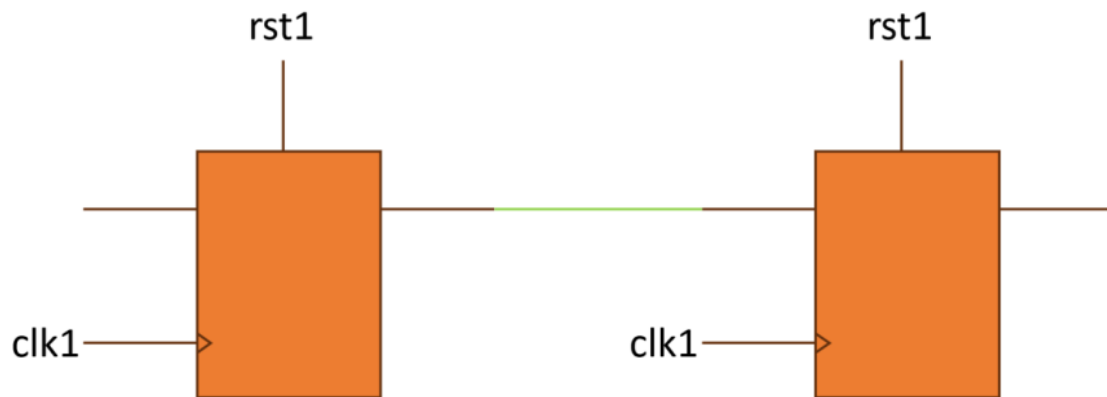


CDC/RDC Analysis and Verification

Part a)

a)



If the reset deassertion is synchronized with the clock, there will be no CDC (Clock Domain Crossing) or RDC (Reset Domain Crossing) problem.

Verilog Implementation

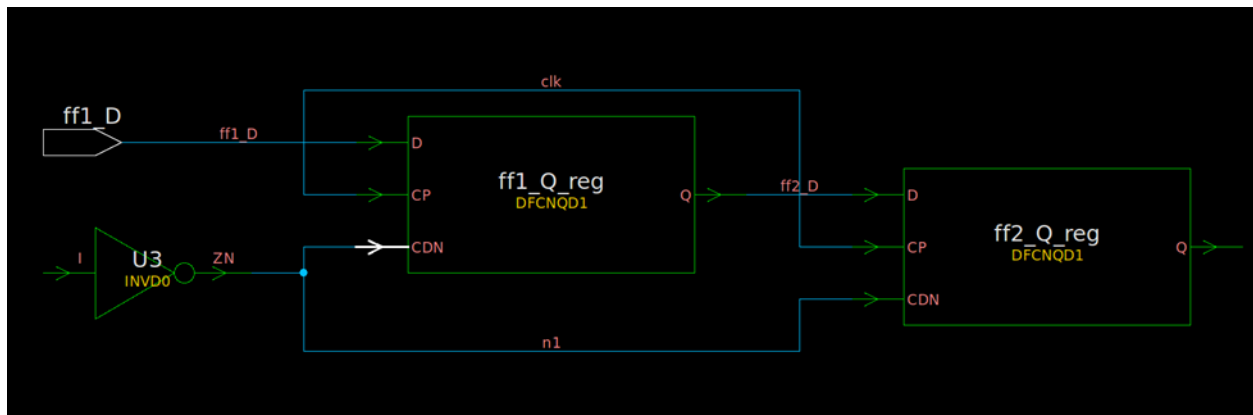
```
module my_module_a (  
    input rst ,  
    input clk,  
    input ff1_D,  
    output reg ff2_Q  
);  
wire ff1_rst , ff2_rst , ff1_clk , ff2_clk;  
reg ff1_Q;  
wire ff2_D;  
assign ff2_D = ff1_Q;  
assign ff1_rst = rst;  
assign ff2_rst = rst;  
assign ff1_clk = clk;  
assign ff2_clk = clk;  
always @(posedge ff1_clk , posedge ff1_rst) begin  
    if (ff1_rst) begin  
        ff1_Q <= 0;  
    end  
end
```

```

    else begin
        ff1_Q <= ff1_D;
    end
end
always @(posedge ff2_clk , posedge ff2_rst) begin
    if (ff2_rst) begin
        ff2_Q <= 0;
    end
    else begin
        ff2_Q <= ff2_D;
    end
end
end
endmodule

```

Schematic:



Part b)

Scenario:

Similar to part (a), except that here reset deassertion is not synchronized with the clock.

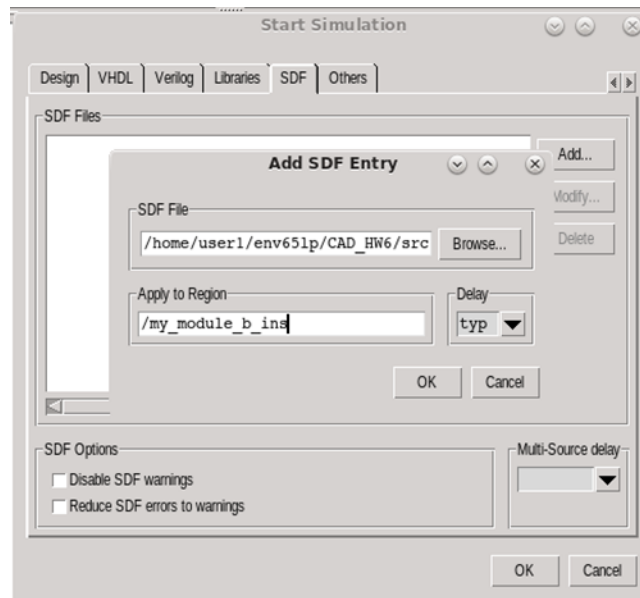
Simulation Procedure:

```

Loading EDA settings ...
[user1@edahost CAD HW6]$ sh run vsim
Running vsim / Quantesim ...

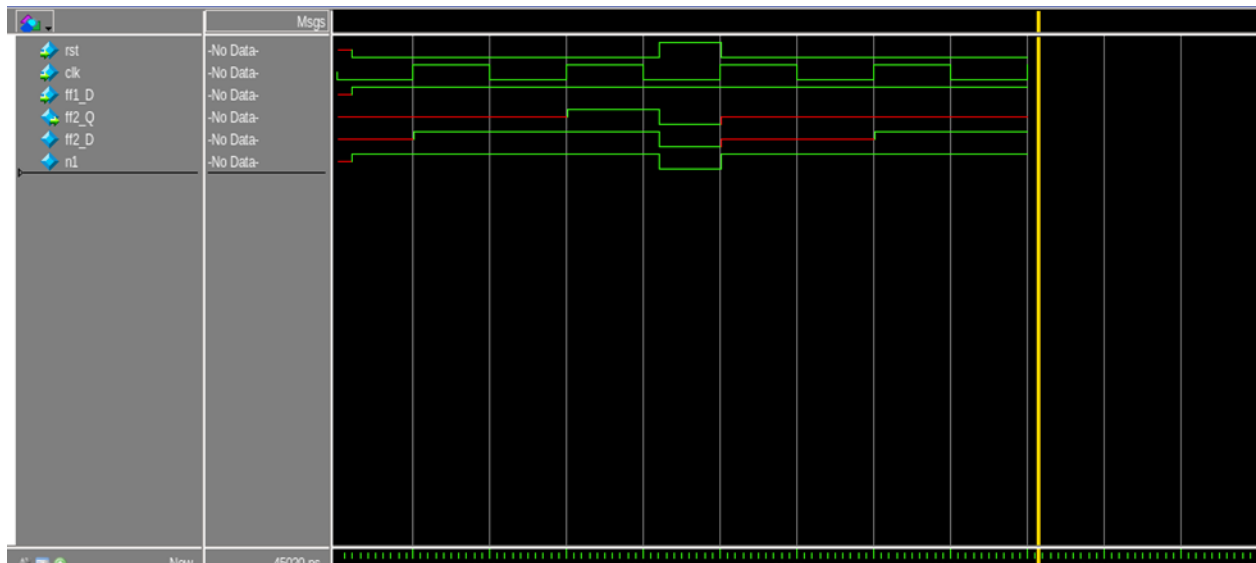
```

The SDF file was added to the simulation



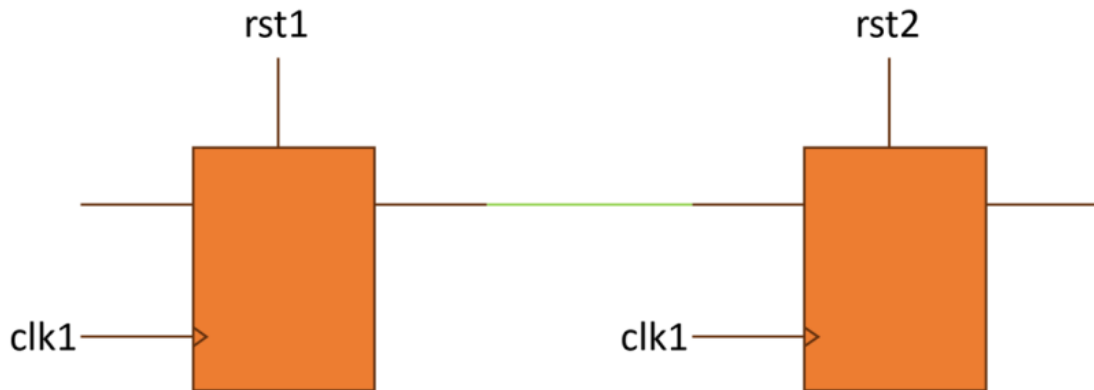
Simulation Result:

It can be observed that because the reset signal was not synchronized with the clock, the circuit experiences metastability during data transition.

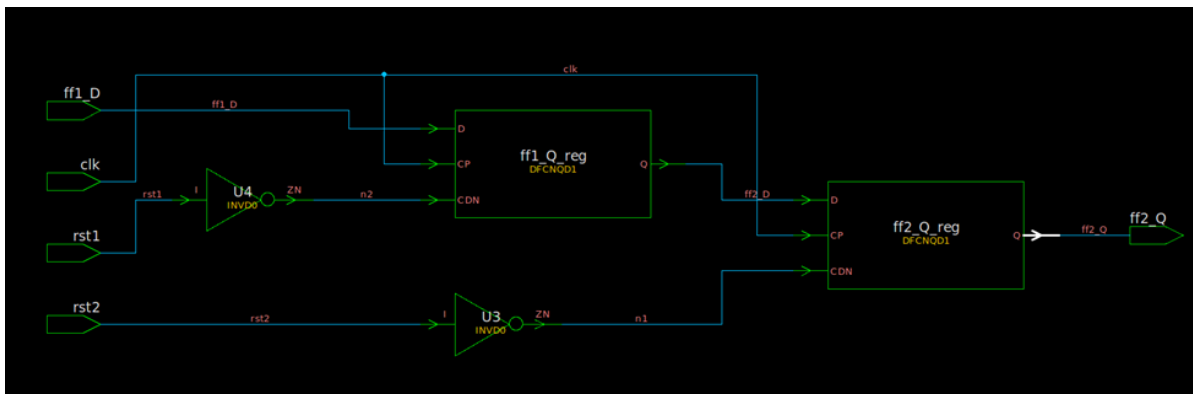


Part c)

c)

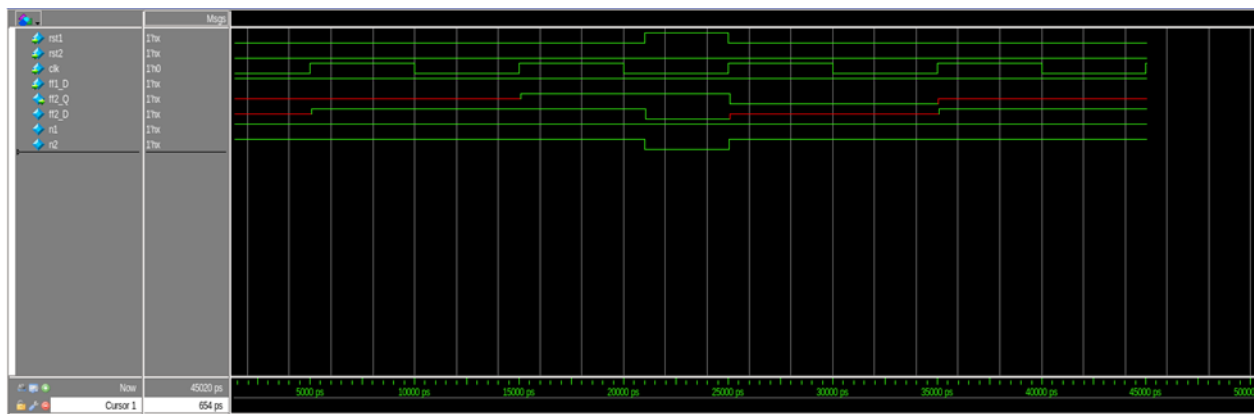


Schematic:



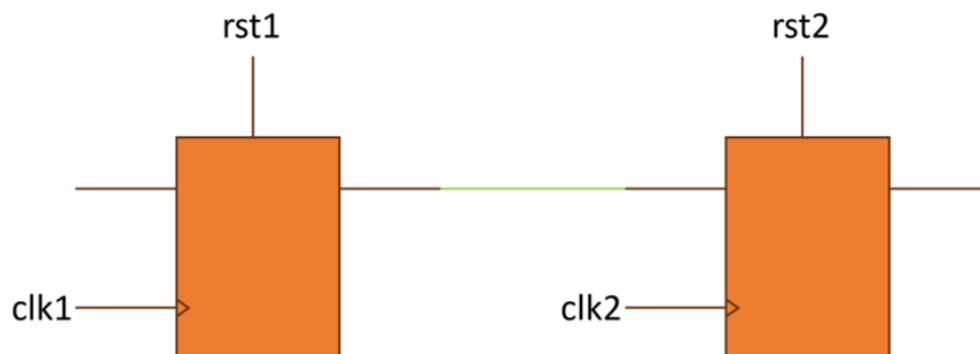
Simulation Result:

The first flip-flop enters a metastable state, which then propagates to the next flip-flop. Hence, metastability occurs and transfers between stages.

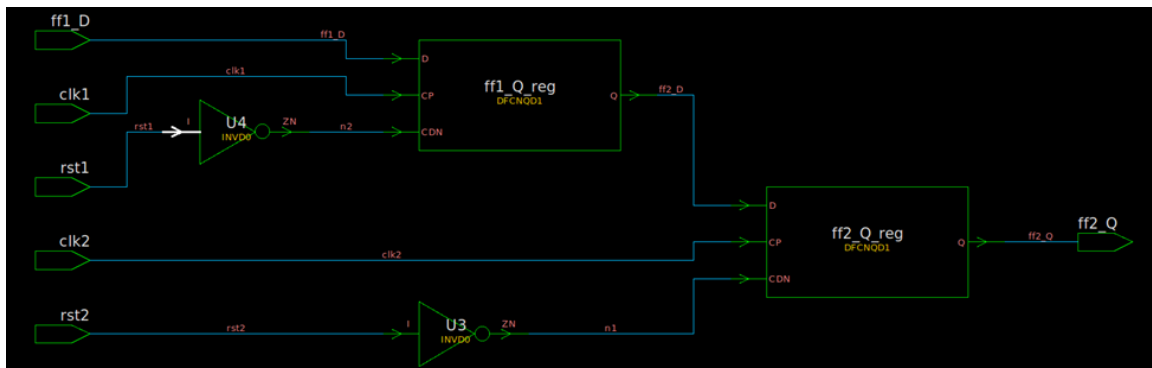


Part d)

d)

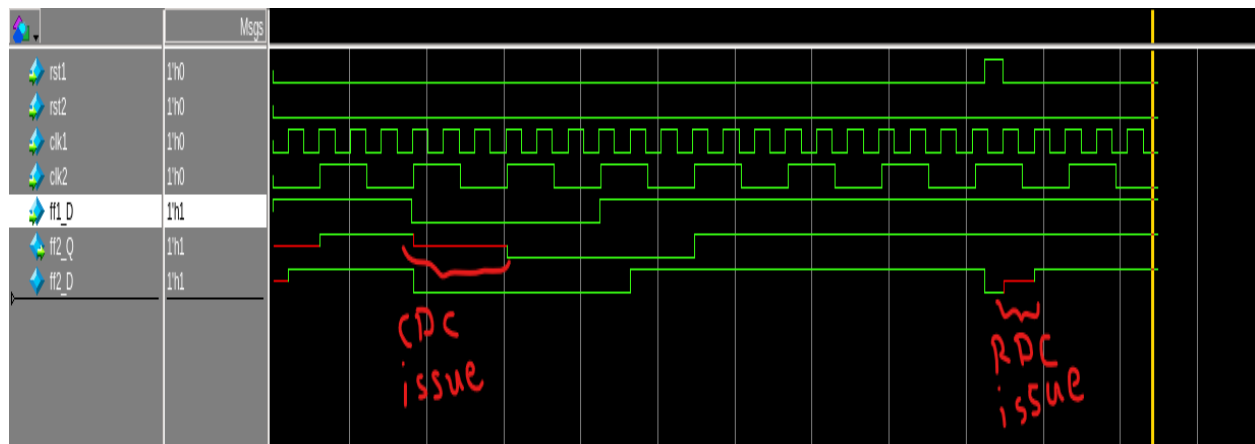


Schematic:



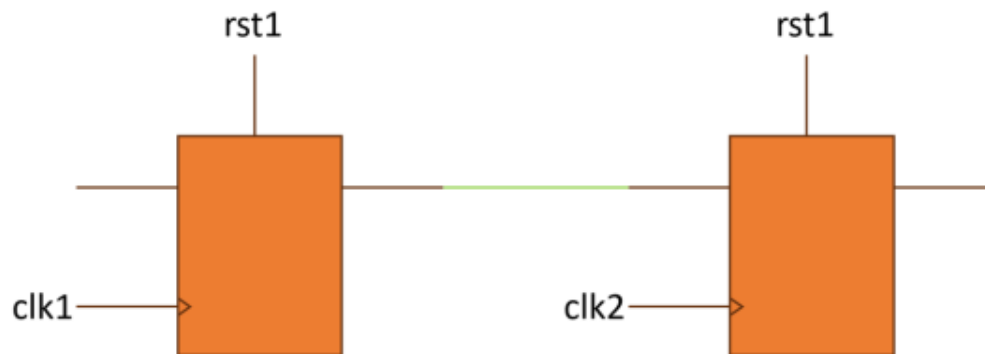
Simulation Results

It can be observed that both CDC and RDC problems are present simultaneously in this circuit.

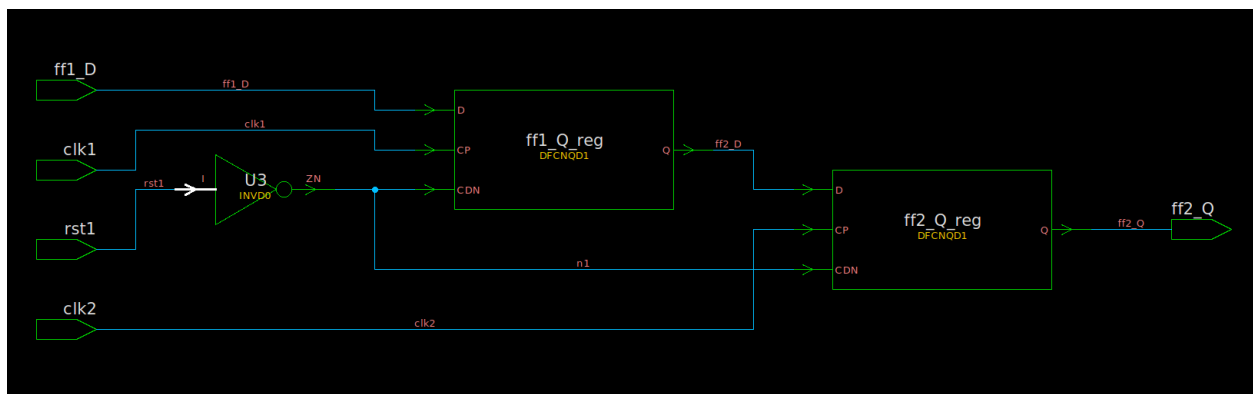


Part e)

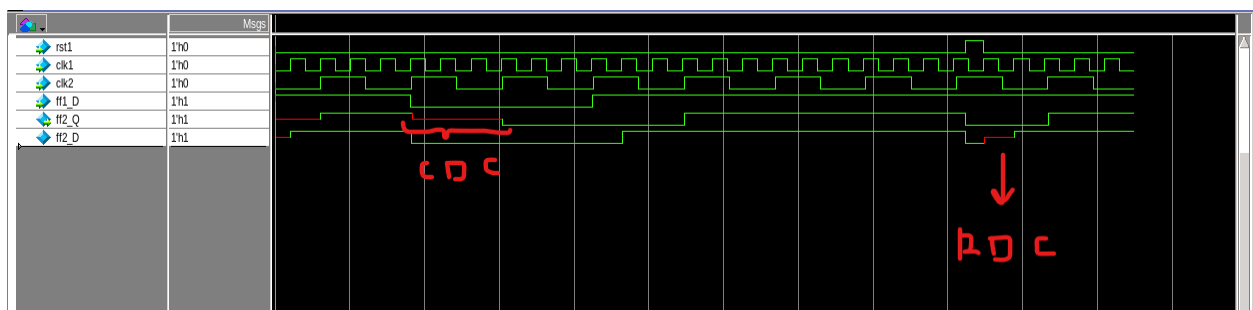
e)



Schematic:

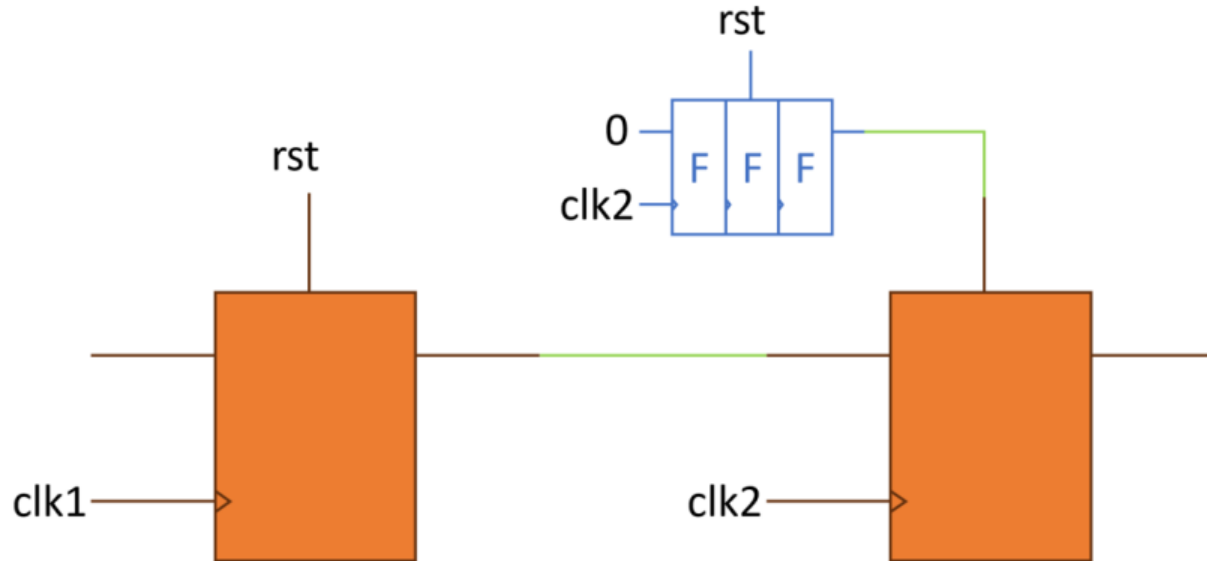


Simulation Result:

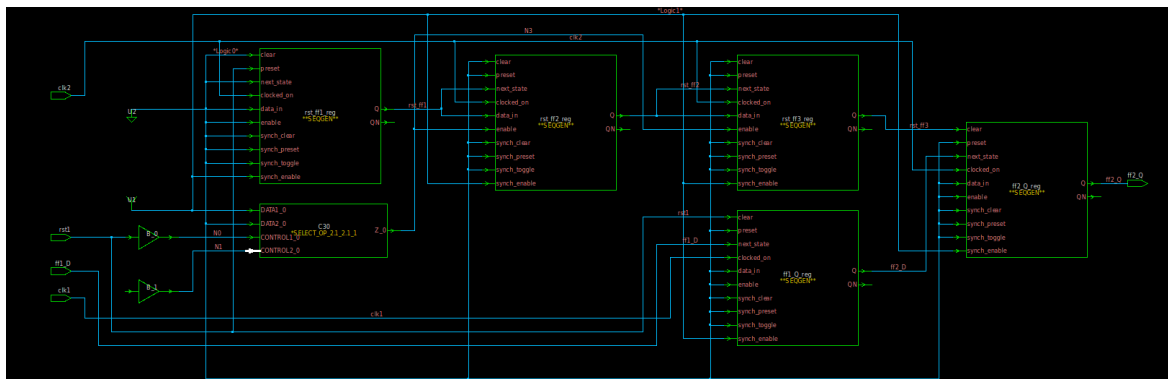


Again, both **Clock Domain Crossing** and **Reset Domain Crossing** issues are visible.

Part f)



Schematic:



This circuit also encounters a full RDC problem.

